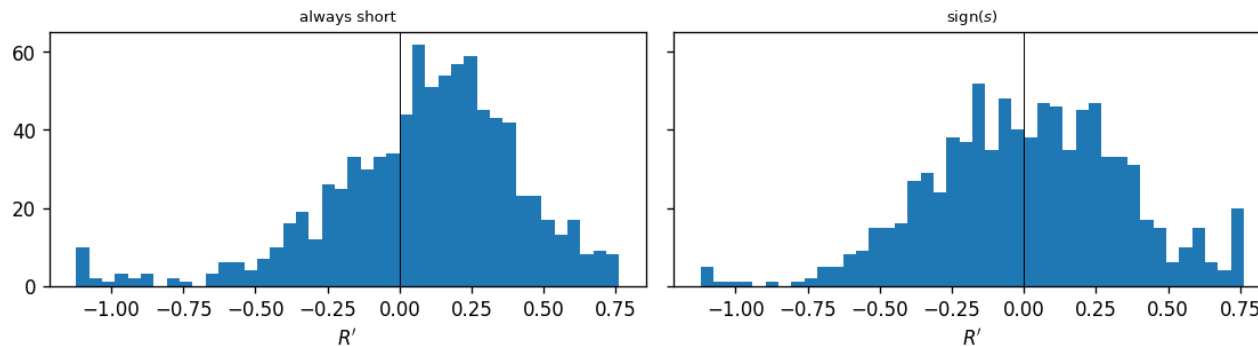


0DTE nearest-OTM straddle: rule table by strategy, every intraday entry window — delta-hedged, held to cash-settle

Rule table by strategy — entry 09:35 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	Sharpe _{clk}	n_{buy}	%buy	Sharpe _{clk} ^x
<i>Short volatility (always short)</i>												
all models (no forecast)	863	0.076	0.385	-3.97	0.90	-2.97	23.37	5.76	3.11	0	0.0	2.44
<i>sign(s) (± 1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	863	0.042	0.391	-3.47	3.97	-0.00	19.95	3.16	1.71	289	33.5	1.04
block-diagonal ridge	863	0.017	0.392	-3.47	3.97	0.23	19.53	1.26	0.68	405	46.9	0.02
block-diagonal ridge, without the FOMC columns	863	0.015	0.393	-3.47	3.97	0.24	19.52	1.14	0.62	408	47.3	-0.04
LightGBM [†]	863	0.037	0.391	-3.47	3.97	0.05	19.83	2.80	1.51	322	37.3	0.85
XGBoost [†]	863	0.040	0.391	-3.47	3.97	0.04	19.89	2.99	1.61	306	35.5	0.95
lasso (causally tuned) [†]	863	0.028	0.392	-3.47	3.97	0.16	19.65	2.09	1.13	372	43.1	0.46
lasso ($\alpha = 10^{-4}$)	863	0.019	0.392	-3.47	3.97	0.22	19.54	1.39	0.75	400	46.3	0.09
elastic net (causally tuned) [†]	863	0.023	0.392	-3.47	3.97	0.11	19.61	1.74	0.94	384	44.5	0.28

block-diagonal ridge, midpoint fills: entry 09:35 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



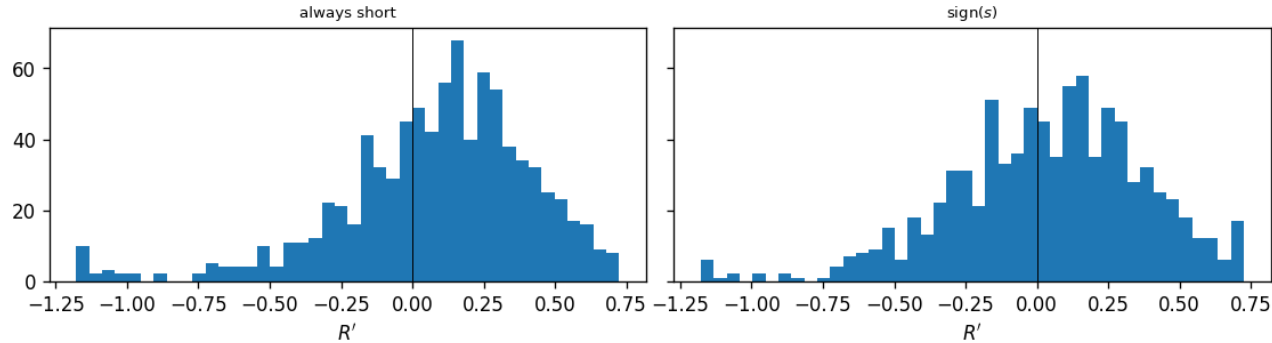
Distribution of the return of the position taken at 09:35 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 09:35 ET, hold those strikes to the official close, and delta-hedge at 10:00 and then every 30 minutes. 863 expiration days (the other 3 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread). **The 09:35 entry.** The first stamp of the session is 09:35, and no forecast bar starts there: the forecasts are for half-hour bars 09:30–10:00, 10:00–10:30, and so on. The forecast used at 09:35 is therefore the one issued at 09:30 for the 09:30–10:00 bar, which is known five minutes before the entry. It covers 09:30–10:00 while the straddle is entered at 09:35; it is used as issued, with no rescaling for the five minutes. The implied side is priced the same way: its diurnal share is the 09:30–10:00 bar's share of the rest of the session, averaged over prior days only (the same expanding mean as at every other clock), and the hours to the close are the chain's, 6.42 hours (6 h 25 min). This design was fixed before the 09:35 numbers were computed.

Rule table by strategy — entry 10:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	864	0.078	0.371	-2.95	0.89	-1.96	9.38	6.15	3.32	0	0.0	2.29
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	864	0.035	0.378	-2.95	1.48	-1.12	7.64	2.75	1.48	307	35.5	0.48
block-diagonal ridge	864	0.039	0.377	-2.95	1.48	-1.13	7.72	3.06	1.66	303	35.1	0.65
block-diagonal ridge, without the FOMC columns	864	0.035	0.378	-2.95	1.48	-1.10	7.62	2.70	1.46	312	36.1	0.45
LightGBM [†]	864	0.048	0.376	-2.95	1.48	-1.16	7.93	3.78	2.04	286	33.1	1.03
XGBoost [†]	864	0.045	0.377	-2.95	1.48	-1.16	7.84	3.48	1.88	281	32.5	0.87
lasso (causally tuned) [†]	864	0.031	0.378	-2.95	1.48	-0.96	7.51	2.44	1.32	310	35.9	0.31
lasso ($\alpha = 10^{-4}$)	864	0.038	0.377	-2.95	1.48	-1.11	7.68	2.95	1.59	294	34.0	0.59
elastic net (causally tuned) [†]	864	0.032	0.378	-2.95	1.48	-1.08	7.57	2.52	1.36	314	36.3	0.36

block-diagonal ridge, midpoint fills: entry 10:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



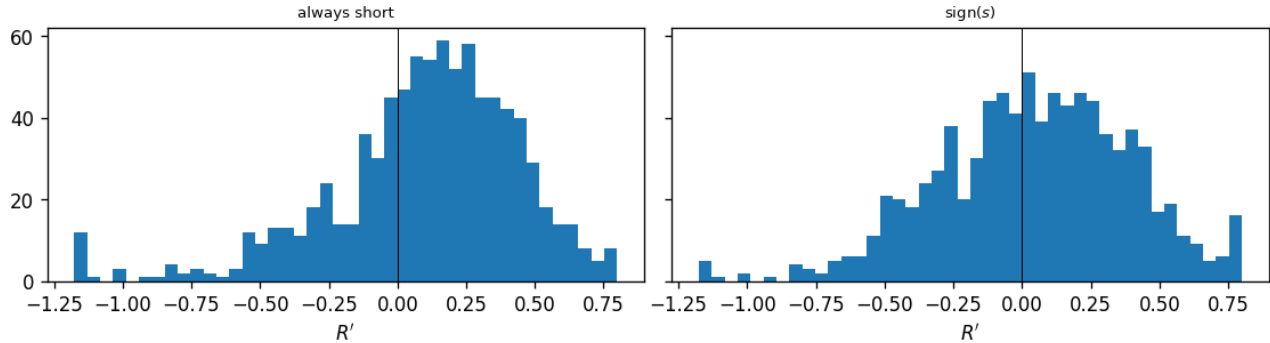
Distribution of the return of the position taken at 10:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 10:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 864 expiration days (the other 2 of the deck’s 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 10:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	865	0.096	0.366	-2.00	0.88	-1.44	4.42	7.74	4.18	0	0.0	3.51
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	865	0.036	0.377	-1.81	2.00	-0.22	2.66	2.81	1.52	289	33.4	0.87
block-diagonal ridge	865	0.046	0.376	-1.81	2.00	-0.02	2.61	3.56	1.92	301	34.8	1.27
block-diagonal ridge, without the FOMC columns	865	0.049	0.376	-1.81	2.00	-0.01	2.62	3.82	2.06	311	36.0	1.41
LightGBM [†]	865	0.043	0.376	-1.81	2.00	-0.07	2.63	3.35	1.81	295	34.1	1.16
XGBoost [†]	865	0.049	0.376	-1.81	2.00	-0.09	2.66	3.80	2.05	287	33.2	1.40
lasso (causally tuned) [†]	865	0.044	0.376	-1.81	2.00	-0.12	2.66	3.43	1.85	310	35.8	1.20
lasso ($\alpha = 10^{-4}$)	865	0.051	0.375	-1.81	2.00	-0.05	2.65	4.01	2.16	287	33.2	1.51
elastic net (causally tuned) [†]	865	0.046	0.376	-1.81	2.00	-0.13	2.67	3.62	1.95	297	34.3	1.30

block-diagonal ridge, midpoint fills: entry 10:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



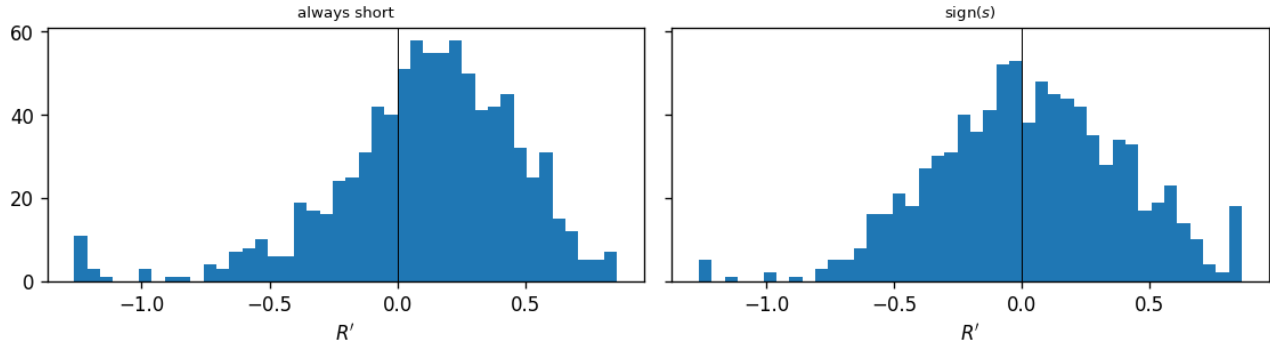
Distribution of the return of the position taken at 10:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 10:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 865 expiration days (the other 1 of the deck’s 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 11:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	865	0.103	0.388	-2.30	0.87	-1.57	5.53	7.78	4.20	0	0.0	3.58
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	865	0.044	0.399	-2.09	2.30	0.19	3.36	3.25	1.75	364	42.1	1.15
block-diagonal ridge	865	0.033	0.400	-2.09	2.30	0.14	3.36	2.46	1.33	365	42.2	0.73
block-diagonal ridge, without the FOMC columns	865	0.029	0.401	-2.09	2.30	0.14	3.35	2.10	1.13	378	43.7	0.53
LightGBM [†]	865	0.030	0.401	-2.09	2.30	0.18	3.34	2.21	1.19	362	41.8	0.59
XGBoost [†]	865	0.038	0.400	-2.09	2.30	0.19	3.35	2.80	1.51	370	42.8	0.91
lasso (causally tuned) [†]	865	0.033	0.400	-2.09	2.30	0.14	3.36	2.45	1.32	379	43.8	0.72
lasso ($\alpha = 10^{-4}$)	865	0.034	0.400	-2.09	2.30	0.13	3.36	2.47	1.33	362	41.8	0.73
elastic net (causally tuned) [†]	865	0.033	0.400	-2.09	2.30	0.12	3.36	2.45	1.32	374	43.2	0.72

block-diagonal ridge, midpoint fills: entry 11:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



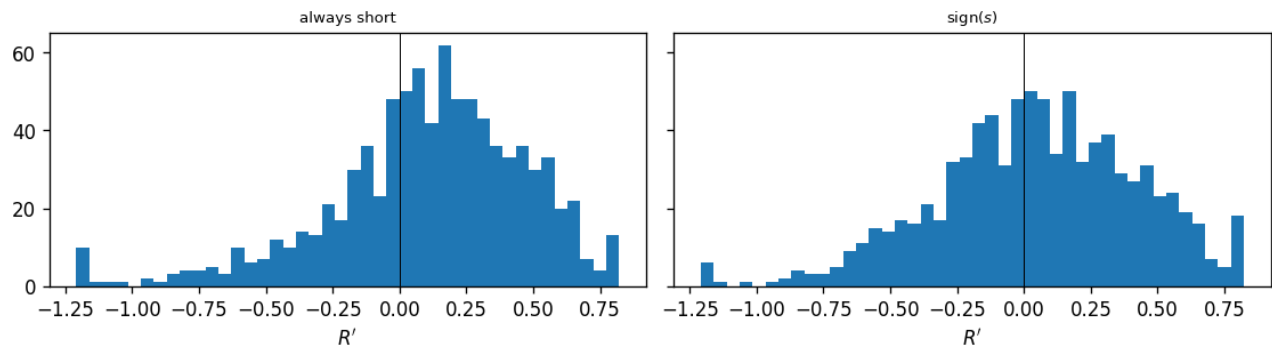
Distribution of the return of the position taken at 11:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 11:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 865 expiration days (the other 1 of the deck’s 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 11:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	865	0.101	0.407	-3.05	0.89	-1.81	8.19	7.32	3.95	0	0.0	3.33
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	865	0.046	0.416	-2.31	3.05	0.05	5.77	3.26	1.76	333	38.5	1.15
block-diagonal ridge	865	0.052	0.416	-2.31	3.05	0.04	5.81	3.65	1.97	336	38.8	1.36
block-diagonal ridge, without the FOMC columns	865	0.050	0.416	-2.31	3.05	0.24	5.70	3.56	1.92	349	40.3	1.31
LightGBM [†]	865	0.039	0.417	-2.31	3.05	0.06	5.73	2.74	1.48	333	38.5	0.87
XGBoost [†]	865	0.042	0.417	-2.31	3.05	0.04	5.75	2.95	1.59	334	38.6	0.99
lasso (causally tuned) [†]	865	0.041	0.417	-2.31	3.05	0.08	5.73	2.92	1.58	344	39.8	0.97
lasso ($\alpha = 10^{-4}$)	865	0.041	0.417	-2.31	3.05	0.07	5.73	2.87	1.55	351	40.6	0.94
elastic net (causally tuned) [†]	865	0.047	0.416	-2.31	3.05	0.07	5.77	3.31	1.78	347	40.1	1.18

block-diagonal ridge, midpoint fills: entry 11:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



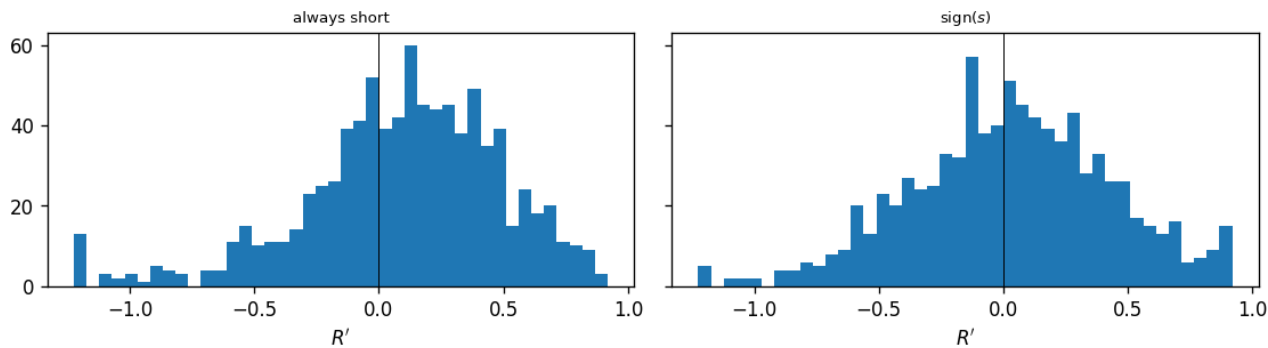
Distribution of the return of the position taken at 11:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 11:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 865 expiration days (the other 1 of the deck’s 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 12:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.087	0.434	-2.97	0.93	-1.48	5.45	5.88	3.17	0	0.0	2.58
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.025	0.442	-2.97	1.98	-0.67	4.12	1.68	0.91	364	42.0	0.33
block-diagonal ridge	866	0.033	0.442	-2.06	2.97	0.26	3.91	2.22	1.20	367	42.4	0.62
block-diagonal ridge, without the FOMC columns	866	0.030	0.442	-2.06	2.97	0.27	3.90	2.03	1.09	379	43.8	0.51
LightGBM [†]	866	0.040	0.441	-2.10	2.97	0.09	3.97	2.67	1.44	360	41.6	0.86
XGBoost [†]	866	0.023	0.442	-2.06	2.97	0.07	3.95	1.52	0.82	372	43.0	0.24
lasso (causally tuned) [†]	866	0.025	0.442	-2.97	1.98	-0.60	4.10	1.64	0.88	385	44.5	0.30
lasso ($\alpha = 10^{-4}$)	866	0.036	0.441	-2.06	2.97	0.22	3.92	2.39	1.29	372	43.0	0.71
elastic net (causally tuned) [†]	866	0.025	0.442	-2.10	2.97	0.08	3.95	1.67	0.90	382	44.1	0.32

block-diagonal ridge, midpoint fills: entry 12:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



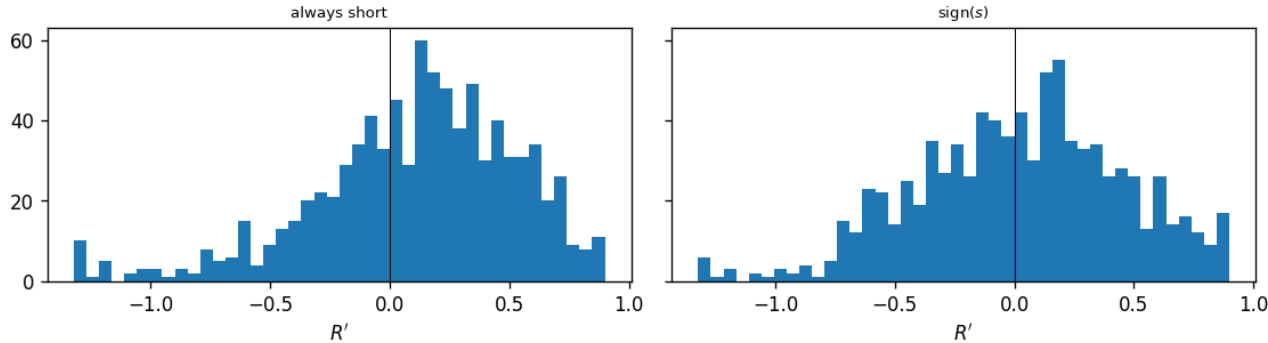
Distribution of the return of the position taken at 12:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 12:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 12:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.097	0.466	-4.31	0.93	-1.88	10.76	6.12	3.30	0	0.0	2.72
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.021	0.475	-2.37	4.31	0.59	8.36	1.31	0.71	379	43.8	0.14
block-diagonal ridge	866	0.027	0.475	-2.37	4.31	0.53	8.36	1.69	0.91	369	42.6	0.34
block-diagonal ridge, without the FOMC columns	866	0.024	0.475	-2.37	4.31	0.61	8.35	1.47	0.79	395	45.6	0.22
LightGBM [†]	866	0.024	0.475	-2.37	4.31	0.63	8.34	1.48	0.80	385	44.5	0.23
XGBoost [†]	866	0.024	0.475	-2.37	4.31	0.61	8.35	1.47	0.80	391	45.2	0.22
lasso (causally tuned) [†]	866	0.023	0.475	-2.37	4.31	0.56	8.36	1.44	0.78	406	46.9	0.20
lasso ($\alpha = 10^{-4}$)	866	0.024	0.475	-2.37	4.31	0.53	8.36	1.50	0.81	371	42.8	0.24
elastic net (causally tuned) [†]	866	0.027	0.475	-2.37	4.31	0.54	8.36	1.67	0.90	389	44.9	0.33

block-diagonal ridge, midpoint fills: entry 12:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



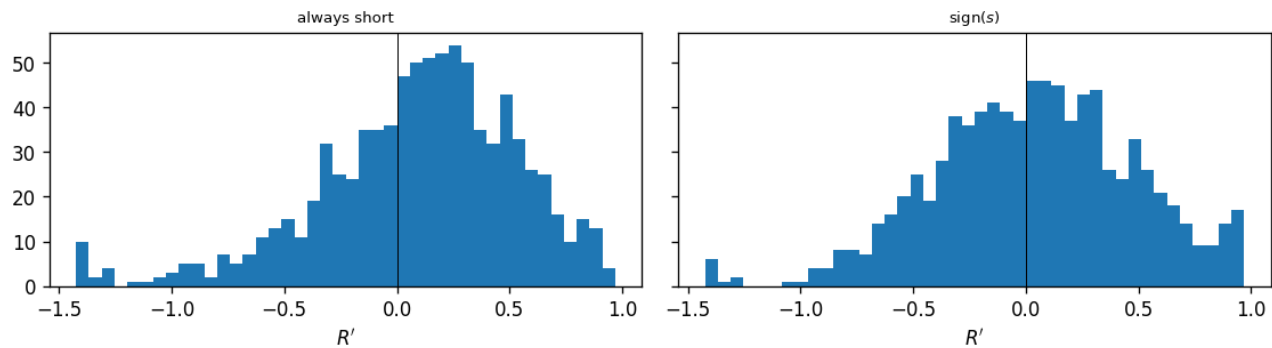
Distribution of the return of the position taken at 12:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 12:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 13:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.083	0.498	-5.47	0.95	-2.48	19.28	4.93	2.66	0	0.0	2.09
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.026	0.504	-2.44	5.47	1.38	16.47	1.51	0.82	446	51.5	0.26
block-diagonal ridge	866	0.044	0.503	-2.44	5.47	1.15	16.52	2.55	1.37	418	48.3	0.81
block-diagonal ridge, without the FOMC columns	866	0.037	0.503	-2.44	5.47	1.18	16.50	2.17	1.17	435	50.2	0.61
LightGBM [†]	866	0.027	0.504	-2.44	5.47	1.36	16.47	1.59	0.86	447	51.6	0.30
XGBoost [†]	866	0.029	0.504	-2.44	5.47	1.35	16.47	1.68	0.91	452	52.2	0.35
lasso (causally tuned) [†]	866	0.029	0.504	-2.44	5.47	1.19	16.50	1.69	0.91	435	50.2	0.35
lasso ($\alpha = 10^{-4}$)	866	0.039	0.503	-2.44	5.47	1.17	16.50	2.27	1.23	421	48.6	0.66
elastic net (causally tuned) [†]	866	0.025	0.504	-2.44	5.47	1.20	16.51	1.49	0.80	438	50.6	0.24

block-diagonal ridge, midpoint fills: entry 13:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



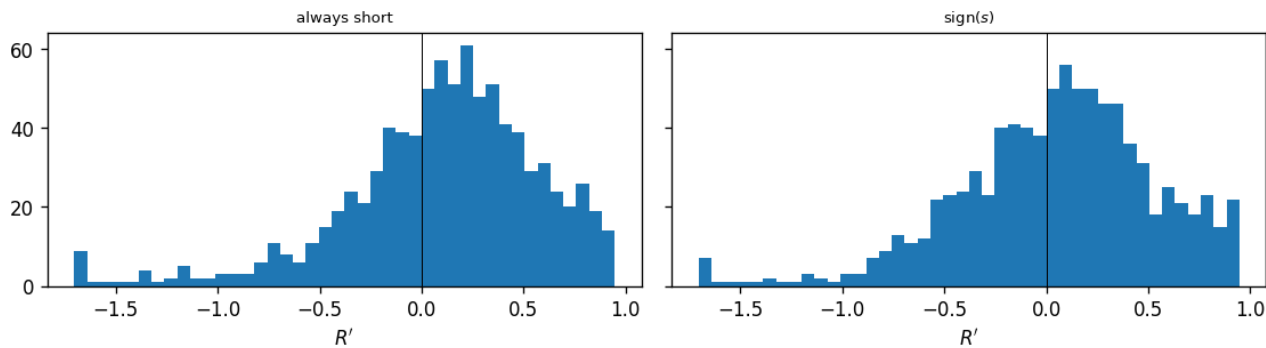
Distribution of the return of the position taken at 13:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 13:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 13:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.088	0.557	-6.72	1.00	-3.24	28.71	4.65	2.51	0	0.0	1.97
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.057	0.561	-2.77	6.72	1.78	25.16	2.99	1.61	306	35.3	1.08
block-diagonal ridge	866	0.058	0.561	-3.14	6.72	1.32	25.36	3.02	1.63	270	31.2	1.09
block-diagonal ridge, without the FOMC columns	866	0.052	0.561	-3.14	6.72	1.35	25.30	2.72	1.47	292	33.7	0.93
LightGBM [†]	866	0.043	0.562	-3.14	6.72	1.43	25.22	2.27	1.22	300	34.6	0.69
XGBoost [†]	866	0.056	0.561	-3.14	6.72	1.43	25.30	2.93	1.58	303	35.0	1.05
lasso (causally tuned) [†]	866	0.048	0.562	-3.14	6.72	1.40	25.25	2.50	1.35	302	34.9	0.81
lasso ($\alpha = 10^{-4}$)	866	0.055	0.561	-3.14	6.72	1.33	25.33	2.89	1.56	267	30.8	1.02
elastic net (causally tuned) [†]	866	0.050	0.562	-3.14	6.72	1.38	25.27	2.62	1.42	295	34.1	0.88

block-diagonal ridge, midpoint fills: entry 13:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



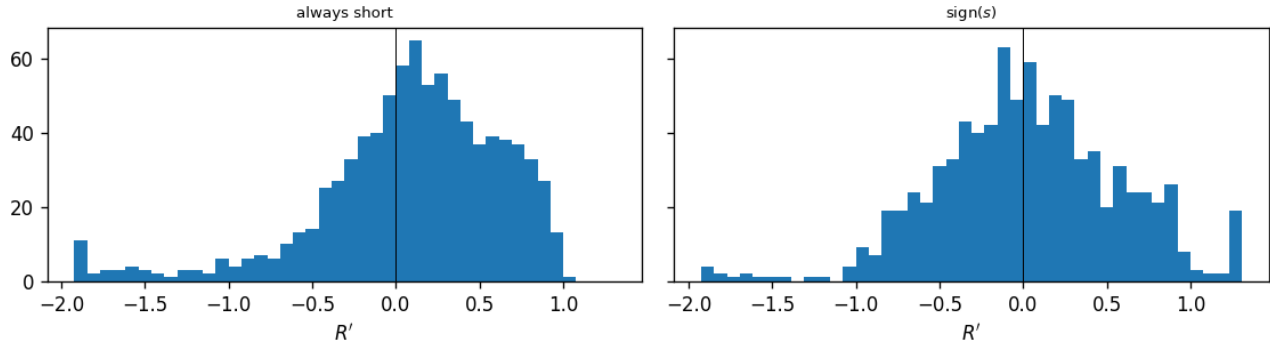
Distribution of the return of the position taken at 13:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 13:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 14:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.080	0.623	-6.94	1.00	-2.74	20.45	3.77	2.04	0	0.0	1.36
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.026	0.628	-2.93	6.94	1.45	18.25	1.22	0.66	451	52.1	-0.01
block-diagonal ridge	866	0.041	0.627	-2.93	6.94	1.41	18.22	1.93	1.04	427	49.3	0.38
block-diagonal ridge, without the FOMC columns	866	0.034	0.627	-2.76	6.94	1.64	18.18	1.59	0.86	442	51.0	0.19
LightGBM [†]	866	0.039	0.627	-2.93	6.94	1.61	18.17	1.83	0.99	469	54.2	0.33
XGBoost [†]	866	0.032	0.627	-2.93	6.94	1.63	18.19	1.51	0.82	472	54.5	0.15
lasso (causally tuned) [†]	866	0.022	0.628	-2.93	6.94	1.39	18.28	1.02	0.55	450	52.0	-0.11
lasso ($\alpha = 10^{-4}$)	866	0.046	0.627	-2.93	6.94	1.41	18.21	2.16	1.16	424	49.0	0.50
elastic net (causally tuned) [†]	866	0.020	0.628	-2.93	6.94	1.42	18.29	0.93	0.50	446	51.5	-0.16

block-diagonal ridge, midpoint fills: entry 14:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



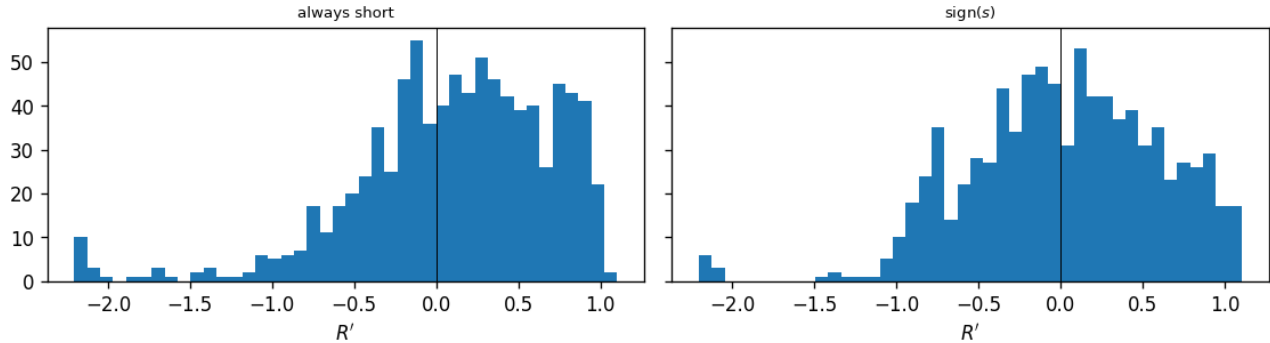
Distribution of the return of the position taken at 14:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 14:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 14:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.091	0.718	-8.43	1.05	-3.71	32.65	3.71	2.00	0	0.0	1.50
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.035	0.723	-6.46	8.43	1.07	29.74	1.41	0.76	335	38.7	0.26
block-diagonal ridge	866	0.031	0.723	-6.46	8.43	0.66	29.81	1.25	0.67	342	39.5	0.17
block-diagonal ridge, without the FOMC columns	866	0.033	0.723	-6.46	8.43	0.66	29.82	1.36	0.73	350	40.4	0.23
LightGBM [†]	866	0.045	0.722	-6.46	8.43	1.06	29.78	1.82	0.98	350	40.4	0.48
XGBoost [†]	866	0.043	0.722	-6.46	8.43	1.12	29.75	1.74	0.94	367	42.4	0.44
lasso (causally tuned) [†]	866	0.031	0.723	-6.46	8.43	1.15	29.72	1.27	0.68	361	41.7	0.18
lasso ($\alpha = 10^{-4}$)	866	0.021	0.723	-8.43	2.98	-2.97	30.22	0.87	0.47	337	38.9	-0.03
elastic net (causally tuned) [†]	866	0.042	0.722	-6.46	8.43	1.13	29.75	1.69	0.91	358	41.3	0.41

block-diagonal ridge, midpoint fills: entry 14:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



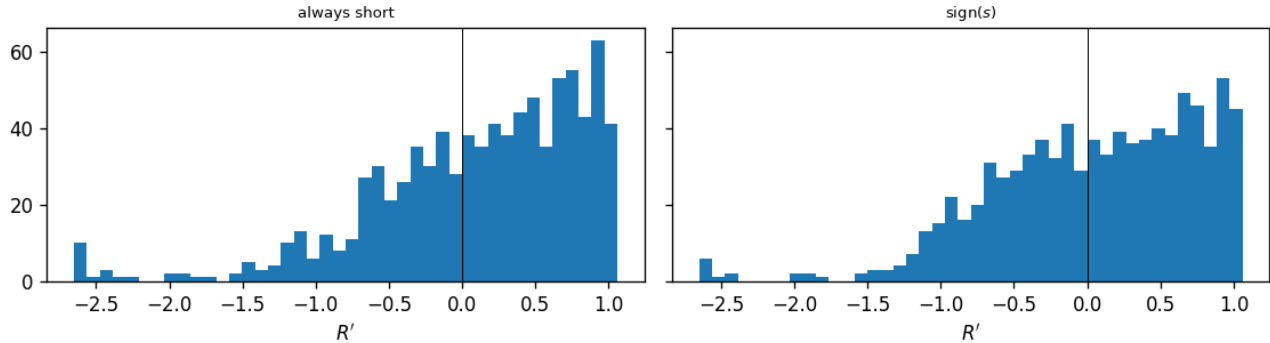
Distribution of the return of the position taken at 14:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 14:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 15:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.095	0.778	-5.81	1.04	-1.95	7.83	3.59	1.93	0	0.0	1.40
<i>sign(s) (±1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.030	0.783	-5.81	3.95	-1.25	6.88	1.12	0.60	201	23.2	0.07
block-diagonal ridge	866	0.065	0.781	-5.81	3.95	-0.97	7.10	2.47	1.33	186	21.5	0.80
block-diagonal ridge, without the FOMC columns	866	0.068	0.781	-5.81	3.95	-0.97	7.11	2.55	1.37	189	21.8	0.84
LightGBM [†]	866	0.053	0.782	-5.81	3.95	-0.94	7.00	2.01	1.08	205	23.7	0.55
XGBoost [†]	866	0.043	0.783	-5.81	3.95	-0.94	6.92	1.60	0.86	220	25.4	0.33
lasso (causally tuned) [†]	866	0.064	0.781	-5.81	3.95	-1.01	7.09	2.40	1.29	198	22.9	0.76
lasso ($\alpha = 10^{-4}$)	866	0.071	0.781	-5.81	3.95	-0.99	7.15	2.67	1.44	185	21.4	0.90
elastic net (causally tuned) [†]	866	0.047	0.782	-5.81	3.95	-1.12	7.00	1.78	0.96	190	21.9	0.43

block-diagonal ridge, midpoint fills: entry 15:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



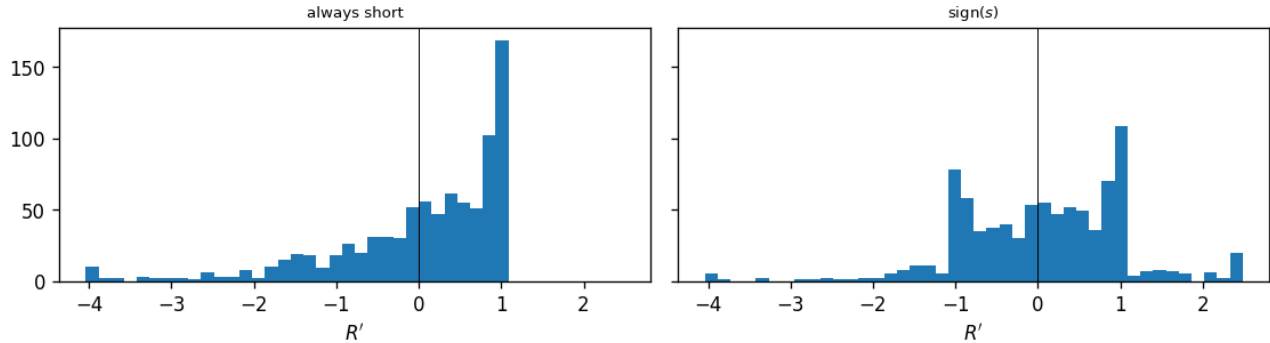
Distribution of the return of the position taken at 15:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 15:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 15:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.017	1.115	-9.38	1.08	-2.26	9.18	0.44	0.24	0	0.0	-0.23
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.064	1.114	-6.37	9.38	0.12	9.07	1.70	0.92	286	33.0	0.44
block-diagonal ridge	866	0.091	1.112	-6.37	9.38	0.62	8.95	2.41	1.30	346	40.0	0.83
block-diagonal ridge, without the FOMC columns	866	0.092	1.112	-4.45	9.38	1.01	8.82	2.45	1.32	355	41.0	0.85
LightGBM [†]	866	0.104	1.111	-6.37	9.38	0.62	8.96	2.76	1.49	290	33.5	1.01
XGBoost [†]	866	0.102	1.111	-6.37	9.38	0.52	9.00	2.72	1.46	279	32.2	0.99
lasso (causally tuned) [†]	866	0.103	1.111	-6.37	9.38	0.40	9.04	2.74	1.48	321	37.1	1.00
lasso ($\alpha = 10^{-4}$)	866	0.105	1.111	-6.37	9.38	0.95	8.84	2.79	1.51	358	41.3	1.03
elastic net (causally tuned) [†]	866	0.075	1.113	-6.37	9.38	0.27	9.05	1.98	1.07	307	35.5	0.60

block-diagonal ridge, midpoint fills: entry 15:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



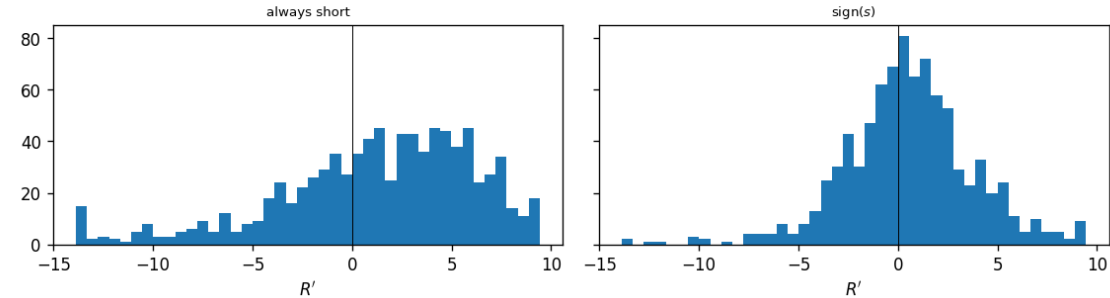
Distribution of the return of the position taken at 15:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window; the mass at -1 is the days the straddle expires worthless).

One trade a day: enter the nearest-OTM straddle at 15:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

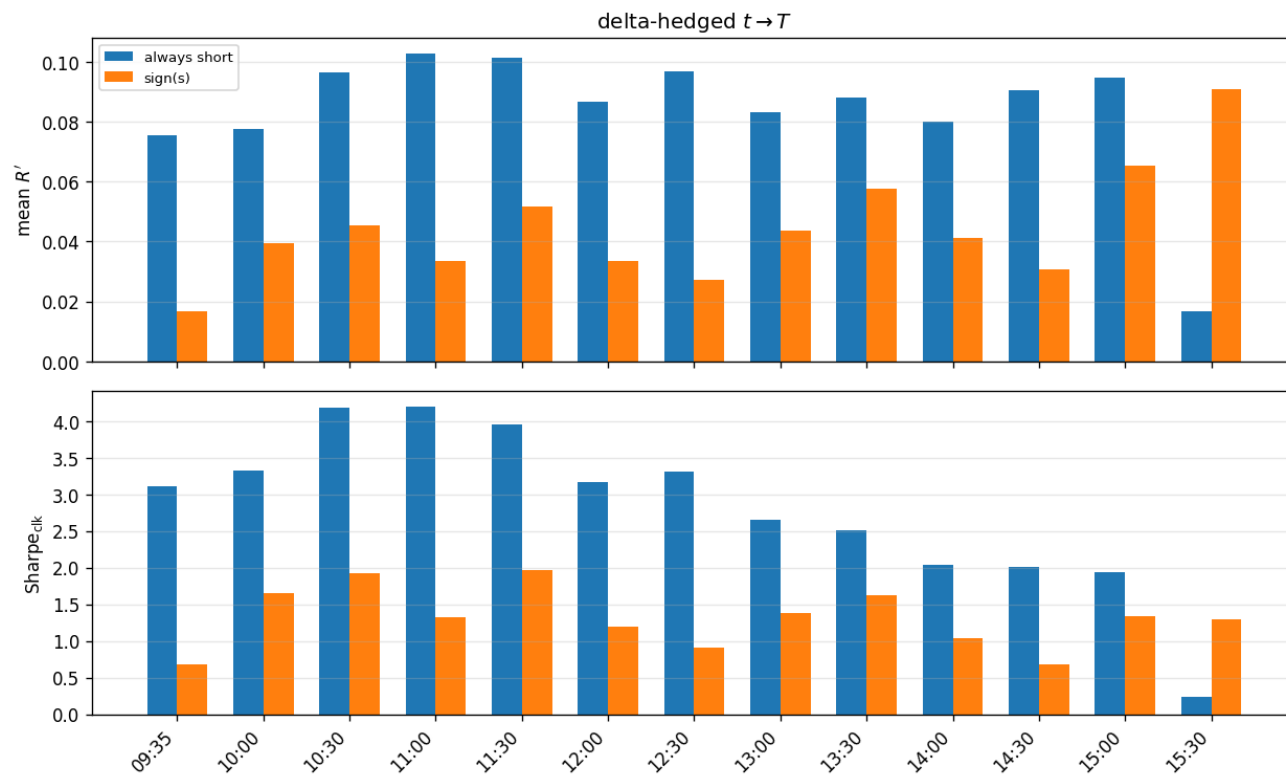
Rule table by strategy — pooled over the thirteen entry windows

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	<i>n</i> _{buy}	%buy	Sharpe ^x _{ann}
<i>Short volatility (always short)</i>												
all models (no forecast)	866	1.090	5.746	-58.68	10.56	-2.51	16.34	5.58	3.01	0	0.0	2.28
<i>sign(s) (±1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.487	3.798	-17.72	52.74	2.69	42.06	3.78	2.04	295	34.1	0.93
block-diagonal ridge	866	0.576	3.874	-24.00	58.68	3.49	60.11	4.38	2.36	292	33.7	1.28
block-diagonal ridge, without the FOMC columns	866	0.548	3.828	-15.43	58.68	3.92	61.57	4.21	2.27	320	37.0	1.17
LightGBM [†]	866	0.573	3.888	-24.00	58.68	3.55	59.29	4.34	2.34	305	35.2	1.26
XGBoost [†]	866	0.564	3.904	-24.00	58.68	3.52	58.32	4.25	2.29	303	35.0	1.22
lasso (causally tuned) [†]	866	0.522	3.799	-24.00	52.74	2.61	43.15	4.04	2.18	320	37.0	1.07
lasso ($\alpha = 10^{-4}$)	866	0.579	3.635	-24.00	41.81	1.20	21.73	4.69	2.53	290	33.5	1.37
elastic net (causally tuned) [†]	866	0.493	3.900	-24.00	58.68	3.37	58.82	3.72	2.01	309	35.7	0.93

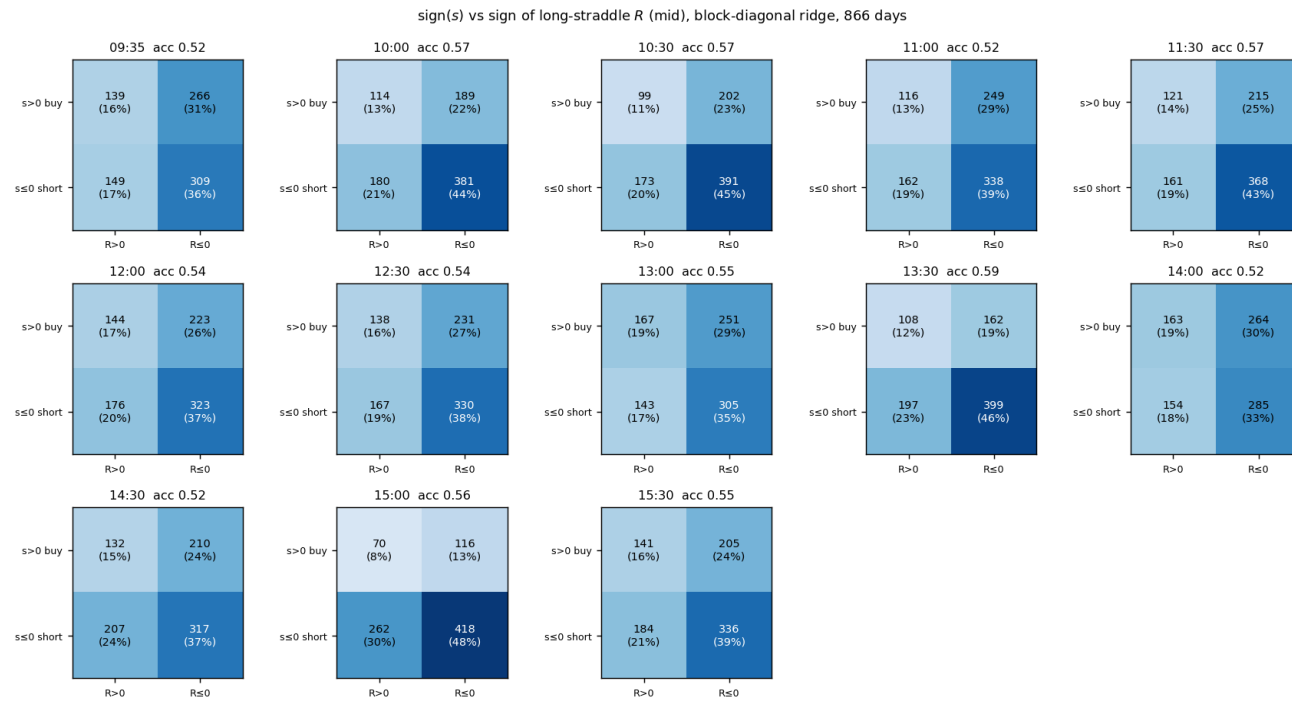
block-diagonal ridge, midpoint fills: daily sum over the thirteen entry windows 09:35-15:30 ET; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



Distribution of the return of the daily sum over the thirteen entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' and $\text{Sharpe}_{\text{clk}}$ by entry clock (grouped bars). $\text{Sharpe}_{\text{clk}}$ is that clock's daily series $\times \sqrt{252}$, not the pooled $\text{Sharpe}_{\text{ann}}$. Delta-hedged $t \rightarrow T$: hold entry K to official close, rebalance Δ every 30 minutes (from 09:35 the first rebalance is at 10:00).



Confusion matrix per entry clock: $\text{sign}(s)$ (buy if $s > 0$) against the sign of the long-straddle midpoint return. 15:30 is the last-30-min trade.

Each row is the daily sum of the thirteen overlapping remaining-session books: enter at each clock 09:35–15:30 ET, hold those strikes to the official close, delta-hedge every 30 minutes (from 09:35 the first re-hedge is at 10:00). 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Notes

Straddle here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean}/\text{std}$, with no autocorrelation correction. “ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of that clock’s series: one remaining-session short per expiration day (enter at this clock, hold those strikes to the close, delta-hedge every 30 minutes). It answers “if I only entered at this clock, what is my annual Sharpe?” It is *not* the book’s $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of $R_d^{\text{day}} = \sum_t R'_{d,t}$, the sum of the day’s thirteen overlapping remaining-session books. That reserved name is the book’s daily P&L after pooling the session. The thirteen books overlap; they are not thirteen successive holds. $\text{Sharpe}^{\times}$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the thirteen positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook’s column. The other column this table adds to the deck’s is $\text{Sharpe}_{\text{ann}}^{\times}$, the same rule’s annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck’s.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}_{\text{hr}}^2 h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day’s remaining-session share of realized variance, $RV_t / \sum_{s \geq t} RV_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck’s treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat. At the 09:35 entry the forecast is the one issued at 09:30 for the 09:30–10:00 bar, w_t is that bar’s share, and h_t is the chain’s hours to the close.